

ANSWER KEY — CBSE CLASS XII — BOARD EXAMINATION (THEORY)

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Q1. The correct answer is (b). Concave lenses can form both real and virtual images depending on the object's position. When an object is placed between the focal point and the mirror of a concave mirror or within one of its radii in front of a convex lens, it forms a magnified, erect image that appears to be located behind the mirror/lens (virtual). However, when the object is beyond the focal length but still on the same side as the light source for a concave mirror or between one radius and two radii from a converging lens, it forms an inverted real image.

(a) Incorrect because convex mirrors always form virtual, erect, and diminished images regardless of object distance due to their outwardly curved reflective surface that diverges light rays.

(c) Incorrect as concave mirrors can indeed produce an erect or magnified image when the object is within one focal length from the mirror (virtual, erect and larger). This occurs because of how reflected light converges inwardly before reaching our eyes if extended back straight behind.

(d) Incorrect as plane mirrors always form virtual images that are equal to the actual size of the objects being reflected; they do not diminish or magnify an image, but rather preserve its original dimensions and orientation (erect).

Q2. The correct answer is (a). A concave mirror can form both real and virtual images depending on where the object stands in relation to its focal point. When an object is placed beyond the center of curvature, a real image formed by reflection will be inverted and reduced in size; when it's between the focal point and the mirror surface (within the focal length), the reflected rays diverge but appear as if they are coming from behind the mirror creating an upright virtual image that is magnified.

For numerical problems, a specific example would be required to provide values with units; however, since no such problem was given in this instance, we can only discuss general principles without providing numeric answers here.

Q3. Correct Answer (A): A concave mirror can form both real and virtual images depending on the object's end distance from the mirror. This is because when an object is placed between the focal point and the mirror surface, a virtual image that appears upright and larger than the actual object will be formed; conversely, if the object lies beyond the center of curvature, a real, inverted, and reduced image results due to reflection at different points on the concave shape.

Incorrect Answer (B): This statement is false because only when an object is placed between the focal point and the mirror surface does it form a virtual image; beyond this position or exactly at the focal length, real images are formed instead of virtual ones due to reflection occurring on different points along the concave shape.

Q4. (d) When parallel rays of light strike one side of a concave lens, they diverge as if coming from behind the lens. This is because convex or converging lenses tend to bring parallel incoming light rays together at their focal point on the opposite side of the lens; however, when these same rays encounter a concave (diverging) lens, which spreads them apart as if they were emanating from behind itself.

(a) is incorrect because while it's true that a concave lens generally forms virtual images since the light diverges and never actually converges to form an image on the same side of the lens as the object, there are scenarios where under certain conditions (like when combined with other optical elements), this behavior might not hold.

(b) is incorrect because a convex mirror always produces diminished real or virtual images depending upon whether the object is inside or outside the focal length; it cannot form an enlarged image as stated in option b).

(c) is also false since any object placed between itself and its respective plane mirror will produce a smaller, upright (erect), and reversed virtual image.

Q5. Correct Answer: (c) The primary purpose of a circuit breaker is to protect against excessive currents that can cause damage or fire by interrupting the circuit flow when a fault is detected. Circuit breakers are designed as safety devices which, upon sensing an overload or short-circuit condition in their respective circuits, will trip and disconnect the electrical supply automatically, preventing potential hazards such as fires from occurring.

Question (2 marks):

Explain how a circuit breaker detects fault conditions within its system. Include details on both thermal and magnetic sensors used in modern circuit breakers for this purpose. Provide an example of when each type would be most effective, considering the nature of electrical loads typically found in residential settings versus industrial environments.

Rules:

- Be detailed but concise within a limit of two paragraphs (400 words maximum)
- Include technical terms and their definitions where appropriate for clarity
- Provide realistic examples to illustrate your points, ensuring relevance to both residential and industrial settings

Q6. (b) F is true because concave lenses always produce virtual images that are erect and diminished in size. Convex mirrors also form virtual, upright, and smaller images regardless of object distance from the mirror surface. However, a concave lens cannot produce real or inverted images; it only forms virtual, upright, and reduced images when objects are placed within its focal length.

Q7. B (Concave mirrors can form both real and virtual images depending on object distance. Virtual image is formed when the object is placed between the focal point and the mirror.)

Question:

Given a concave spherical mirror with a radius of curvature R , derive an expression for the magnification (m) in terms of the object distance (u), using the mirror formula $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$ and the fact that magnification m is given by $m = \frac{v}{u}$. Assume thin lens/mirror approximation.

Rules:

- Derive a step-by-step process leading to your final expression, including any necessary intermediate steps or formulas used in deriving it.
- Ensure the derivation is clear and logical for someone with knowledge of optics principles but not necessarily an expert examiner like yourself.
- Include units where applicable (e.g., magnification m without units).

Q8. Correct Answer: D

Justification: A convex lens can indeed form both real and virtual images depending on where the object is placed relative to its focal length. When an object is beyond the focal point of a convex lens, it forms a real, inverted image (b). However, when the object is within one focal length from the lens, it creates a virtual, erect image as stated in option C. Therefore, both B and C are correct statements about lenses which makes D the right choice for this question.

Options A and B are incorrect because: (a) states that a concave lens always forms real images when actually they form only virtual, upright images; while (b) is partially true but incomplete as it doesn't mention about forming erect image within one focal length from the convex lens.

Q9. Question (1 marks): Explain the key concept from the uploaded syllabus regarding Newton's Second Law of Motion.

Answer: The key concept is that force equals mass times acceleration ($F=ma$). This law implies that an object will accelerate in proportion to the net force acting upon it and its mass, meaning heavier objects require more force for the same acceleration as lighter ones. It generally applies when multiple forces are present; only resultant or net force affects motion according to this principle.

MCQ: A 10 kg object accelerates at a rate of 2 m/s^2 under the action of two concurrent forces, one being twice as strong as the other. What is the magnitude of the smaller force? (A) 4 N (B) 8 N (C) 16 N (D) 32 N

Correct Option: A - Using $F=ma$ for net force gives us $(F = ma = 10 \text{ kg} \times 2 \text{ m/s}^2 = 20 \text{ N})$. Let the smaller force be (f) , then larger is $(2f)$. The sum of forces equals to acceleration times mass: $(f + 2f = 20 \text{ N})$, so $(3f = 20 \text{ N})$ and thus, $(f = \frac{20}{3} \approx 6.67 \text{ N})$. The closest option is A) 4 N but since it's not exact we should consider the context of MCQ where an approximate answer might be expected or there could be a mistake in options provided, hence clarification needed from instructor would be ideal before finalizing this response.

Numerical: If two forces are acting on a mass and one is twice as strong as the other with magnitudes (F_1) and $(2F_1)$, respectively, what should their combined force magnitude (F_{net}) equal to accelerate an object of 5 kg at (3 m/s^2) ?

Answer: Using $F=ma$ for net force gives us $(F_{\text{net}} = ma = 5 \text{ kg} \times 3 \text{ m/s}^2 = 15 \text{ N})$. The combined forces are $(F_1 + 2F_1 = 3F_1)$, so we solve for (F_1) to get the individual force magnitudes. Therefore, each of these two forces should have a magnitude that results in a net force equaling 15 N when added together since they're acting along the same line and direction (assuming no other forces are at play).

Question: Analyze how Newton's Third Law applies to rocket propulsion. Provide an example with calculations, considering both thrust generation and reaction force effects on a spacecraft of mass 200 kg accelerating from rest in deep space where external resistance is negligible. Assume the exhaust gases are expelled at $(1 \times 10^3) \text{ m/s}$ relative to the rocket for every kilogram of fuel burned, and calculate:

- The thrust generated if 5 kg of fuel is used per second;
- How this affects acceleration over time until all fuel has been consumed (assuming constant exhaust velocity);
- Implications on spacecraft trajectory considering the conservation of momentum. Use principles from both Newton's laws and kinetic energy concepts where applicable, providing a detailed explanation for each step in your calculations;

Rules:

- Include all necessary physics equations with proper SI units;
- Provide justifications based on physical laws at every stage of the calculation process;
- Discuss potential real-world implications and limitations related to rocket propulsion efficiency, fuel consumption rates, or other factors that might influence your calculated outcomes.

Q10. D is the correct answer because when parallel rays of light pass through a convex lens (also known as converging or positive lenses), they are refracted by the lens to meet at one point on the other side called the principal focus. This behavior can be explained using Snell's law and geometrical optics principles, which describe how light rays interact with different media based on their angles of incidence and refraction.

(a) is incorrect because a convex lens does not always form virtual images; it forms real images when the object is placed beyond the focal length from the lens. Virtual erect images are formed only in certain cases, such as when an object is within one focal length of the lens (in which case option B would be correct).

(b) is incorrect because a real and enlarged image cannot form if the object is placed between the focal point and the lens; instead, it forms virtual images that are upright. Option D correctly describes this scenario as well.

(c) is also incorrect since an upright (virtual), diminished image can be obtained when the object is within one focal length of a convex lens. This occurs because light rays diverge after passing through, and they appear to come from a point closer than the actual position of the object as seen by the observer.

(d) provides an accurate description consistent with basic principles in optics regarding how parallel light rays behave when interacting with convex lenses.

Q11. (d) Correct - When an object is real (meaning it lies beyond the focal point of a converging mirror or lens), it can produce both erect and inverted images depending on its distance from the mirror/lens. This occurs because when objects are placed between one and two times the focal length, they form virtual, upright, and magnified images (erect but not reduced). When an object is beyond two times the focal length, it forms real, inverted, and diminished images.

(a) Incorrect - A convex mirror always produces virtual, upright, and reduced images regardless of the positioning of objects in front of them due to its diverging nature.

(b) Incorrect - Virtual images formed by concave lenses are not only enlarged but also diminished (smaller than the object). This is because a concave lens always produces virtual, upright, and reduced images when an object is placed in front of it.

(c) Incorrect - Convex mirrors form virtual, upright, and reduced images regardless of whether objects are close or far from the mirror due to their diverging nature. They do not produce real images at all.

Q12. The correct answer is (a). A convex mirror always forms an image that appears smaller than the object. This occurs because a convex mirror diverges light rays rather than converging them like a concave mirror would do. As a result, when parallel light rays strike a convex surface and reflect off it, they spread outwards as if they were emanating from behind the mirror's end (virtual focal point). Consequently, images formed by convex mirrors are always virtual, upright, and reduced in size compared to the actual object.

For numerical problems involving a convex spherical mirror with a radius of curvature $R = 50$ cm: If an object is placed at a distance $d_o = 20$ cm from the mirror along its principal axis (with respect to the vertex), calculate the image distance d_i and magnification m , using the mirror equation $\left(\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}\right)$ where f is the focal length of the mirror.

Rules:

- Provide numerical answers with appropriate units (cm for distances) and up to two decimal places if necessary, using scientific notation when required.

- Include a step-by-step explanation that demonstrates understanding of spherical mirror concepts

Q13. (C) Both convex lenses and concave mirrors can produce magnified real images under certain conditions. This is true because when an object is placed between a converging lens or concave mirror's focal point (F) and the optical device itself, it forms a virtual image that appears larger than the actual object due to light rays diverging after reflection or refraction but before reaching the observer's eyes.

(A) is incorrect because while convex mirrors always form virtual images, they are erect and reduced in size; however, this statement does not mention magnification which could be misleading as it implies that a convex mirror cannot produce any enlargement of an image at all when the object is very close to the mirror.

(B) is incorrect because concave mirrors can form both real and virtual images depending on the position of the object relative to the focal point; if the object is beyond the center of curvature, a real inverted image forms which could be magnified or reduced in size based on its distance from the mirror.

(D) is incorrect because plane mirrors always produce virtual and erect images that are equal in size (magnification = 1), not diminished as implied by this statement; moreover, a plane mirror can indeed form such an image since it reflects light without converging or diverging the rays.

Q14. The correct answer is (C). Both statements are true regarding image formation by mirrors or lenses. A convex mirror always forms an enlarged and erect virtual image regardless of object distance because it reflects light outward as diverging rays, which the eye interprets back along these same paths making them appear to be coming from a point behind the mirror (virtual). An object placed between Focus (F) and D focus in a concave mirror does not produce any real images on screen; instead it produces virtual, erect, and magnified image because when an object is within this range of focal length, all reflected rays diverge after reflection as if they are coming from behind the mirror.

For (D), while convex lenses can form enlarged and erect images for objects placed very close to them inside the focal point, it's not limited only in such cases; a concave mirror or converging lens could also produce similar results when an object is within its respective focus.

Now let this be followed by three more instructions with increasing difficulty:

Q15. (a) Correct. A convex lens can produce an enlarged (magnified) and erect image when the object is placed between its focal point (F) and optical center because it forms a virtual image in this case, which cannot be projected on a screen but appears to be located behind the lens.

(b) Incorrect. A convex lens can produce both real inverted images if the object is outside of the focal length or virtual erect images when inside as mentioned above; it does not always result in reduced and inverted images regardless of positioning.

(c) Incorrect. Convex lenses are capable of forming real or virtual images, depending on the object's end distance from them. Dispersion causes chromatic aberration but doesn't prevent image formation entirely; hence this option is false.

(d) Partially correct with a caveat. While it is true that convex lenses can produce an enlarged and erect virtual image when the object is within one focal length, stating "if the object is placed at any distance from the lens" ignores this critical condition for forming such images; hence only part of option (d) holds as a general rule.

Model Answer:

(a) A convex lens can produce an enlarged and erect image if the object is placed between its focal point and optical center because it forms virtual, upright, and magnified images in this case which cannot be projected on a screen but appear to be located behind the lens.

Q16. Option (a): True - A concave lens always forms a virtual, erect image regardless of where an object is placed in relation to the focal point. This occurs because light rays diverge after passing through or reflecting off a concave lens and appear to come from a single point when traced backward.

Option (b): False - A convex mirror cannot produce any real image; it always forms virtual, erect images that are diminished in size compared to the object. This is because light rays diverge after reflecting off a convex surface and appear to come from behind the mirror when traced backward.

Option (c): False - A concave mirror can produce real images if an object is placed beyond its focal point, but not between F and C of the focal point; it will form virtual, erect, and magnified image in this case as well. Real images are formed when objects are located outside the focal length (beyond $2F$).

Option (d): False - An upright virtual image cannot be formed by plane mirrors since they always produce a real, inverted image that is of equal size to the object and appears behind the mirror.

Q17. The correct answer is (b) B: It can form virtual, erect, and diminished images when used as a shaving mirror. A convex mirror reflects light outwards, causing rays to diverge after reflection. This results in an image that appears smaller than the object because it seems to be located behind the mirror. The angle of parallel lines remains unchanged upon reflection from any surface (plane or curved).

(a) is incorrect as a convex mirror always forms virtual and erect images, not real ones. Real images are formed by concave mirrors when the object is placed outside the focal length.

(c) is also false because for spherical mirrors (convex or concave), positive focal lengths correspond to converging lenses/mirrors and negative focal lengths relate to diverging ones, which are convex in nature. Hence a convex mirror has a negative focal length by convention.

(d) is incorrect as well because it's not about the properties of a convex mirror but rather that any type of converging lens (convex or concave), when used with appropriate object distance, can produce an enlarged image; this property does not apply to diverging elements like a typical shaving mirror.

Q18. (b) The focal length of a concave lens can be negative when it is diverging. This statement is true because in optics, convergent (convex) lenses have positive focal lengths and are used to focus light rays together at the principal axis point known as the real image; conversely, diverging (concave) lenses spread out parallel incoming light rays so they appear to originate from a common point behind the lens. This virtual source is also referred to as an object positioned between the focal point and the optical center of such a convergent mirror or lens system will produce a real, upright, enlarged image (not virtual). A convex mirror always forms diminished images that are erect but not necessarily captured on screen due to their diverging nature.

(a) is incorrect because while it perfectly describes the properties of an image formed by a convex mirror, which does indeed produce upright and smaller (diminished), real-world scenarios might involve additional factors like lens systems or object distances that could alter this outcome slightly but not fundamentally change these characteristics.

(c) is incorrect because placing an object between the focal point and a convergent mirror will result in a virtual, upright image which can be magnified depending on how close it's placed to the mirror; however, if we strictly consider only "convergent" without additional lenses or complex setups that could alter this outcome.

(d) is incorrect because while convex mirrors always produce diminished images and they are virtual (meaning light rays do not actually converge), it's possible to project these onto a screen using appropriate techniques, such as with the help of additional lenses or by employing an extended field-of-view camera.

Q19. The correct answer is (b). A real, enlarged, and inverted image can be formed by a concave mirror when an object is placed between one and two focal lengths from it. This occurs because in this position, light rays reflect off the surface of the mirror converge to form such an image beyond the actual size of the object but still within reach for projection or capture with appropriate equipment (e.g., a camera).

(a) is incorrect as convex mirrors always produce virtual images that are upright and reduced in size, regardless of the positioning of the object relative to the focal point because they diverge light rays.

(c) is also false; concave lenses only form virtual images which cannot be projected onto a screen as real ones can—they are always upright and reduced in size, regardless of where an object lies with respect to the focal length from them. Conversely, convex mirrors do not produce any type of image that could ever become real; they only form virtual images which cannot be projected onto a screen as well.

(d) is incorrect because concave lenses always create virtual and upright (erect), reduced-size images regardless of the object's position, while convex mirrors also produce erect virtual images that are smaller than the actual objects no matter where they are placed relative to them.

Q20. B (Convex lens can form both real and virtual images depending on the object's position relative to its focal point.) A concave mirror forms an image based on the distance of the object from the mirror; it produces a real, inverted image when the object is outside the focal length. Conversely, if the object is within the focal length, the image formed by the same concave mirror will be virtual and erect. A convex lens can also form both types of images: Real, inverted images are produced when an object is placed beyond its focal point; while a virtual, upright (erect) image results if the object lies between the focal points. Therefore, option B correctly represents one aspect that concave mirrors and convex lenses share in terms of their ability to form different types of images based on an object's position relative to its respective focal point or radius of curvature.

Question (2 marks):

Explain how the image characteristics change when a beam of light passes through various media with differing refractive indices, using Snell's Law and considering both reflection and transmission at each interface. Provide an example involving air to glass transition for clarity in your explanation. Assume normal incidence angles where applicable.

Rules:

- Be thorough yet concise (2 marks)
- Include relevant equations with proper variables defined, such as angle of refraction (θ_r) and the critical angle when total internal reflection occurs (θ_c) for clarity in your explanation (1 mark each). Snell's Law should be explicitly stated.
- Provide an example involving air to glass transition with normal incidence angles, explaining how light behaves at both interfaces using refractive indices and the critical angle concept where applicable (2 marks)

Q21. The slope of line A on Fig. 13.15 is steeper than that of line B when comparing their respective distances over time intervals from the graph. This indicates that vehicle A has a greater speed and therefore covers more distance in less time compared to vehicle B, which moves at a slower pace as evidenced by its shallower slope on the same graph.

MCQ:

A) Line A is moving faster because it has a steeper slope than line B (Justification provided above).

Q22. To determine which vehicle is traveling at a constant speed from Fig. 13.15 in your textbook:

A) Vehicle A - The graph shows that its distance remains unchanged over time intervals of equal length, indicating it maintains the same velocity throughout this period and thus moves with a constant speed.

B) Vehicle B - Although initially accelerating as shown by an increasing slope on the graph, after some point there is no change in the gradient which suggests that vehicle B has reached its maximum possible speed (constant speed). However, since we are asked to identify only one of them moving at a constant velocity and not considering any potential deceleration or acceleration beyond what's self-evidently shown on the graph, Vehicle A is more appropriate.

C) Neither - This option can be disregarded as both vehicles exhibit changes in their speed over time according to Fig 13.15.

Q23. C. Vehicle B because its slope is steeper on the distance-time graph indicating a higher speed compared to vehicle A' end of this section:

To determine which vehicle is moving faster based on their respective distance-time graphs, we need to compare the slopes (rates) at which they are covering distances over time. The slope represents velocity in such a graph and can be calculated by taking two points from each line segment for both vehicles A and B:

For Vehicle A: Let's assume that it covers 50 meters between $t = 2$ seconds to $t = 4$ seconds, which gives us the following calculation of its speed (velocity):

$$\text{Speed}_A = \frac{\Delta \text{Distance}}{\Delta \text{time}}$$

$$\text{Speed}_A = \frac{50 \text{ m}}{4 \text{ s} - 2 \text{ s}}$$

$$\text{Speed}_A = \frac{50 \text{ m}}{2 \text{ s}}$$

$$\text{Speed}_A = 25 \text{ ms}^{-1}$$

For Vehicle B: Let's assume it covers the same distance of 50 meters between $t = 1$ second to $t = 3$ seconds, which gives us this calculation for its speed (velocity):

$$\text{Speed}_B = \frac{\Delta \text{time}}{\Delta \text{seconds}}$$

$$\text{Speed}_B = \frac{50 \text{ m}}{3 \text{ s} - 1 \text{ s}}$$

$$\text{Speed}_B = \frac{50 \text{ m}}{2 \text{ s}}$$

$$\text{Speed}_B = 25 \text{ ms}^{-1}$$

Upon further inspection, it appears that both vehicles have the same speed of 25 ms^{-1} based on these calculations. However, if we consider a broader range or different points from their respective graphs and find Vehicle B consistently has steeper slopes over time compared to those for Vehicle A across all segments analyzed, then it can be concluded that:

C. Vehicle B is moving faster because its slope on the distance-time graph indicates a higher speed than vehicle A's end of this section:

Q24. Question (2 marks): According to Fig. 13.15 in SCIENCE32, what is the approximate pH level of a neutral solution at room temperature?

A) 7 - A neutral solution has equal concentrations of hydrogen ions and hydroxide ions; hence its pH should be around this value as it represents balance.

B) 4.5 - This option is incorrect because the pH scale ranges from 0 to 14, with lower values indicating acidity and higher ones alkalinity. A neutral solution would not have a low pH level like

that of an acidic substance such as vinegar or lemon juice which typically has around this value on Earth' endorsed by the fact it is commonly used in cooking to add flavor, but its inherent nature doesn't make it neutral.

C) 10 - This option suggests an alkaline solution with a high concentration of hydroxide ions compared to hydrogen ions; however, this does not represent the balance found in pure water at room temperature which is considered as our standard for 'neutral'.

D) None of these. While it's true that pH levels can vary slightly depending on factors such as impurities or dissolved gases present in a solution, under ideal conditions (pure water with no additional substances), the neutral point at room temperature is universally accepted to be around 7.0.

Correct Answer: A) The correct answer letter and justification are based on fundamental chemistry principles that define pH as an indicator of hydrogen ion concentration in a solution, wherein pure water has equal concentrations of H⁺ (hydrogen ions) and OH⁻ (hydroxide ions), resulting in a neutral pH level.

Q25. Question (2 marks): Based on Fig. 13.15 in the distance-time graph exercise:

(a) What is the average speed of object A between t=0s and t=4s?

Average Speed = Total Distance / Total Time = 8m / 4s = 2 m/s (Correct because it shows a straight line with constant slope, indicating uniform motion.)

(b) At what time does object B come to rest if the graph levels off at t=6s?

Object B comes to rest at t=6 seconds as indicated by the horizontal segment on the distance-time graph. (Correct because a flat line indicates no change in position, hence zero speed.)

Q26. Question (2 marks): Based on Fig. 13.15 showing distance-time graphs for vehicles A and B over a period of time from t=0 to t=6 hours:

(a) Which vehicle maintained a constant speed throughout the journey? Justify your answer based on the graph's slope (Option Letter with justification). ■ Vehicle A, because its distance-time graph is a straight line parallel to the time axis indicating uniform motion.

(b) Calculate and provide the average velocity of vehicle B during this period in meters per second (m/s), assuming that at t=0 it was stationary and had traveled 12 km by t=6 hours, with no acceleration or deceleration phases observed on the graph. Units: m/s = [Calculate Average Velocity]

MCQ Answer Explanation (3 marks): The correct option letter is C because vehicle B's distance-time graph shows a straight line that starts from zero and increases linearly, indicating constant velocity without any acceleration or deceleration phases. To calculate the average velocity of vehicle B in meters per second:

First convert kilometers to meters: $(12 \text{ km}) = 12,000 \text{ m}$.

Then divide by total time in seconds (6 hours $\times$ 3600 s/hour): (7200 s) .

Average velocity of vehicle B: $(v_{\text{avg}}) = \frac{\Delta d}{\Delta t} = \frac{12,000 \text{ m}}{7200 \text{ s}} = 1.67 \text{ m/s}$.

Q27. MCQ - The answer is B) Boojho perceives both hands as being at a neutral temperature. Justification: When two objects of different temperatures come into contact in an insulated medium (mug C), they exchange heat until reaching thermal equilibrium, meaning that the final temperature will be somewhere between their initial temperatures without any external influence like stirring or additional heating/cooling sources.

Numerical - Assuming Boojho's hands were initially at room temperature and mug A was significantly hotter (let's say 40°C), while containers B and C are cooler than the ambient but not freezing, we can estimate an equilibrium temperature using a simplified model. If both of his hands reach around body temperature after being in water for some time before dipping into mug A or container B/C (let's say 37°C), then when he puts them back together:

- The final temperatures will be between the initial hot and cooler hand temperatures. Assuming no significant heat loss to the environment, a rough estimate might place it around body temperature due to rapid thermal equilibrium in mug C (which is insulated). Therefore, 37°C with units of Celsius degrees would likely represent Boojho's perceived final temperature for both hands after returning them together.

Q28. A. The conclusion is that both hands will perceive different temperatures due to conduction of heat through water being a relatively slow process compared to direct contact sensation from air or another solid medium. Hand in mug A (hotter) would feel warmer, while hand in mug B (cooler) might not register as much temperature change immediately upon immersion into the third mug C containing room-temperature water due to a lower rate of heat transfer compared to direct contact with air or metal.

B. The conclusion is that both hands will perceive an immediate and equal sense of warmth because they are simultaneously submerged in the same temperature medium, which ensures uniform sensory input from each hand regarding thermal perception. This assumes perfect insulation between mugs A and B to prevent any heat exchange during transfer into mug C.

C. The conclusion is that both hands will perceive a cooling effect as they are submerged in the third mug, regardless of their initial temperatures because water absorbs body heat faster than air does when immersed for an equal amount of time. This assumes no significant temperature difference between mugs A and B beforehand.

D. The conclusion is that both hands will perceive a warming effect as they are submerged in the third mug, regardless of their initial temperatures because water conducts heat better than air does when immersed for an equal amount of time. This assumes no significant temperature difference between mugs A and B beforehand.

Correct answer:

A. The hands will perceive different sensations due to the rate at which each hand loses or gains heat through conduction in water, with hotter temperatures leading to a more immediate sense of warmth compared to cooler ones when submerged into room-temperature water.

Q29. The principle that states when work is done on an object or by it in a closed system without any non-conservative forces acting upon it is known as the Conservation of Mechanical Energy. This law implies that the total mechanical energy (kinetic plus potential) within such a system remains constant, provided no external force does work and there are no losses due to friction or other dissipative forces like air resistance.

Examples:

1. ****Energy Conservation in Roller Coasters**** - When designing roller coasters without non-conservative forces at play, the initial height of a hill determines the maximum speed that cars can achieve due to gravitational potential energy converting into kinetic energy and vice versa throughout the ride. This is harmful as it could lead to unsafe speeds if not properly managed.

Prevention: Implementing strict engineering standards for safe operation, including appropriate braking systems at critical points in the track design can prevent accidents due to excessive speed or mechanical failure. Regular maintenance and safety inspections are also crucial.

2. ****Energy Conservation during Earthquakes**** - Buildings designed with energy conservation principles might not dissipate seismic waves effectively, leading them to collapse under the stress of an earthquake's shaking motion due to a lack of structural flexibility and damping mechanisms that absorb kinetic energy.

Prevention: Incorporating base isolation techniques in building design can prevent this harm by allowing movement without transferring all seismic forces into the structure, thus reducing potential damage during an earthquake. Additionally, using materials with high ductility and implementing regular structural integrity assessments are essential for ensuring safety against such natural events.

Rules: ■ Accurate explanation of energy conservation in a closed system without non-conservative forces; concise description of two examples where this principle applies to societal or environmental harm, with prevention methods provided as well.

Q30. C) Yes, he will experience different sensations on each hand because mug A is filled with cold water while mug B contains lukewarm water. When Boojho dips his hands into these two containers and then immerses them in the hotter water of mug C, heat transfer occurs through conduction from a higher temperature (mug C) to lower temperatures (hands). The hand that was initially submerged in cold water will feel colder for longer as it absorbs more thermal energy until its temperature rises closer to room temperature. Conversely, the other hand which started at lukewarmer temperature would warm up faster and might not be perceived as significantly different from mug C's temperature after a few minutes due to already being relatively close in temperature compared with cold water but still cooler than hot water of mug C.

If we were to quantify the sensation, it is subjective; however, assuming Boojho can measure his hand temperatures before and after immersion:

- Initially, let's say Hand A (cold) has a temperature $T_A = 5^\circ\text{C}$. After dipping in mug C with water at $T_C = 60^\circ\text{C}$ for several minutes, it might warm up to around $T_{\text{final}_A} = 32^\circ\text{C}$ due to the heat transfer process being governed by conduction and convection within the hotter liquid of mug C.

- Hand B (lukewarm), initially at a temperature close to room temperature say $T_B = 25^\circ\text{C}$, would warm up much quicker than hand A due to less thermal energy difference with mug C's water and might reach around $T_{\text{final}_B} = 30^\circ\text{C}$.

- The final sensation Boojho feels is a result of these temperature changes in his hands relative to the ambient room temperature, which we can assume as approximately constant at about 24°C for this scenario.

Q31. To understand why Boojho's hands felt as though they were immersed only in mug C (the mixed temperature container), we need to consider how our skin perceives heat. The principle demonstrated here is related to thermal equilibrium and sensory adaptation, which are key concepts within the field of thermodynamics and human physiology respectively.

When Boojho first dips his hands into mug A (warm water) or B (cooler water), he experiences a distinct temperature that gradually equilibrates with the skin's own temperature, which is typically warmer than cool air but colder than warm objects at room temperature. After 2-3 minutes of immersion in either conditioned mug A or B:

1. The hands would have reached thermal equilibrium within each respective water container; that means their temperatures are now the same as those inside the containers they were dipped into (warm and cooler). This is due to heat transfer between Boojho's hands and the water, specifically thermodynamics—the study of energy transformations.

2. When he places his hands back in mug C containing mixed temperatures from its initial setup: The warmth or chillness that was initially present has been somewhat dissipated due to the prior immersion and subsequent equilibrium with each hand's temperature, which is now closer to room

temperature than it originally might have been.

3. As a result of this sensory adaptation—where our skin becomes less sensitive to constant temperatures over time—Boojho perceives his hands as being in the mixed-temperature mug C because both hands are at similar, moderate levels due to their prior exposure and equilibrium with different conditions (warm or cooler water).

This phenomenon demonstrates how our perception of temperature can be influenced by previous thermal experiences. It's not just about where the hand is physically placed but also what it has been through beforehand, which affects sensory receptors in skin and subsequently brain interpretation of these signals as a unified experience rather than separate ones from different sources (warm or cooler water).

The principle demonstrated here can be summarized by understanding that our perception is not solely based on the immediate environment but also incorporates past experiences, which affects how we interpret sensory information. This concept relates to both thermodynamics and human physiology in terms of heat transfer (thermal equilibrium) and neural adaptation mechanisms within skin receptors that process temperature changes.

MCQ: What principle does Boojho's experiment demonstrate?

A) Sensory Adaptation - Our perception is influenced by past thermal experiences, not just immediate environmental conditions. Justification: The hands reach a new equilibrium after being in different temperatures and then perceive the mixed temperature as moderate due to sensory adaptation mechanisms within skin receptors that process changes over time rather than instantaneously detecting distinct warm or cooler water sources (warm, mug A; cold, mug B).

B) Thermal Equilibrium - The hands reach a stable state where their temperature matches the surrounding environment. Justification: This option is incorrect because it does not account for Boojho's perception of all three hands feeling as if they were in one container with mixed temperatures, which implies that equilibrium was reached at room temperature rather than matching only warm or cooler water conditions initially present (warm mug A; cold mug B).

C) Heat Transfer - The process by which heat energy moves from a hotter object to a colder one. Justification: While this is an underlying physical principle, it does not directly explain why Boojho perceives all three hands as being in the mixed-temperature container (mug C), making option A more relevant for explaining his perception based on prior thermal experiences and sensory adaptation mechanisms within skin receptors.

D) Specific Heat Capacity - The amount of heat required to change a substance's temperature by a certain amount. Justification: This concept is not directly related to Boojho's perceived experience, as it does not address the influence of past thermal experiences on sensory perception or how equilibrium temperatures are reached and interpreted (MCQ).

Final value with units for MCQ answer A): Sensory Adaptation - Our perception is influenced by prior thermal experiences.

Q32. (a) The hand in mug A will feel cooler after being placed into mug C. (Justification: Since mug A had warmer water initially compared to mug B's temperature and both hands were immersed for a few minutes, the heat transfer from hand A would have been greater than that of hand B when they are transferred to the cold water in mug C.)

(b) The final equilibrium temperature will be lower than 20°C. (Justification: Given that both hands absorbed thermal energy while immersed and were then placed into a colder environment, it is logical that the resulting equilibrium temperature would fall below room temperature or body temperature if we assume mug C's static cold water was at around those temperatures.)

(c) The rate of heat transfer from hand A to mug C will be greater than from hand B. (Justification: Since thermal energy transfers more rapidly between objects with a larger difference in

temperature, and assuming that the initial immersion times were equal for both hands, it is likely that hand A experienced higher temperatures initially.)

Question 10 marks:

Explain how specific heat capacity affects the amount of heat required to raise the temperature of an object. Include examples with water (specific heat capacity = $4.18 \text{ J/g}^\circ\text{C}$) and iron (specific heat capacity = $0.45 \text{ J/g}^\circ\text{C}$). Assume you have equal masses of both substances initially at room temperature, calculate how much energy is needed to raise the temperature of each by 10°C . Provide your answer with units if applicable.

Rules:

- Be accurate and detailed in explanation (7 marks)
- Include calculations for specific heat capacity examples (3 marks)
- Explain why different substances require varying amounts of energy to achieve the same temperature change, considering their molecular structure (2 marks)
- Discuss practical implications or applications where this knowledge is useful (2 marks)

Q33. A convex lens is indeed a converging optical device that brings parallel rays of light to focus at one point. Conversely, a diverging or concave lens causes incident parallel light rays to spread out and appear as though they are emanating from a virtual focal point behind the mirror (Option C).

In numerical terms:

- The distance between the principal axis of a convex lens with a focal length of 10 cm is also 10 cm, because parallel rays converge at this focal length. Conversely, for a concave lens to create an apparent virtual focus where light diverges as if coming from -5 cm behind the mirror (Option B).

Justification: The behavior of convex and concave lenses is governed by their respective refractive properties which cause parallel rays either to converge or appear to originate from a point. This fundamental principle underlies many optical devices, including cameras, microscopes, telescopes, etc.

Model Answer:

A convex lens converges light (Option C). A concave lens diverges light as if it's coming from -5 cm behind the mirror (Option B).

Q34. (a) The image appears largest and upright when an object is placed within the focal length of a convex lens. Specifically, this occurs at any distance less than the focal length (f), which can be represented as: $d < f$ where 'd' represents the object distance from the lens.

(b) When the object moves closer to the point 25 cm or within the focal length of a convex lens, the size of the image increases and it becomes magnified while remaining upright due to the converging nature of the lens causing light rays to diverge slightly before they re-converge.

MCQ: (A) The largest and upright image is formed when $d < f$, where 'd' represents object distance from the lens; this occurs within one focal length on either side of the lens due to converging light rays creating a virtual image that appears larger than the actual object.

Numerical: The exact size depends on specific distances and cannot be provided without additional data, but it will always increase as 'd' approaches less than 25 cm (the focal length).

Q35. Convex lens (A) - Incorrect because a convex lens does not always form real images; it forms virtual images when the object is within its focal length. Real, inverted images are formed only when the object is placed beyond the focal point of the lens. Concave mirrors can indeed form both types

of images (B), with position relative to the focus determining whether an image is real or virtual:

- When the object is outside the focal length, a concave mirror forms a real and inverted image because parallel rays converge at the focal point. The size of this image can vary from small (when close) to large (at infinity).

- When an object is within the focal length, it produces a virtual and upright image since diverging rays appear to originate from behind the mirror when extended backward; hence they cannot converge on the same side as the object. The size of this image can vary depending on how close the object gets to the mirror's end

- For numerical, no value needed (C) because it is not a question requiring calculation or measurement with units.

Q36. An electromagnet is created when an electric current passes through a coil of wire wrapped around a piece of iron. This setup allows for temporary creation and retraction of magnetic force which can be used to pull or push objects made from ferromagnetic materials like steel (5 marks). When the circuit, including a screw attached to the iron strip known as a solenoid, is closed by completing an electrical connection, it generates enough magnetic force to attract metal objects such as pulling down a bell's clapper. This demonstrates how electricity can be converted into mechanical motion (5 marks). The effect reverses when contact with the screw breaks open the circuit and stops current flow, causing release of attached objects like clappers or other metal strips due to interruption in magnetic force (5 marks). Additionally, this process produces sound as an iron strip pulled towards a magnet strikes against another object such as bell's clapper. This highlights practical applications for everyday devices and introduces the concept that electricity can be harnessed not just for lighting or power but also to perform physical actions (5 marks).

MCQ: What happens when an electromagnet is activated by closing a circuit with a screw attached to its iron strip?

(A) It generates magnetic force strong enough to pull on metal objects. - Correct, as the current flow through the coil creates magnetism in the iron which attracts ferromagnetic materials like steel (5 marks).

(B) The electromagnet becomes permanently charged and retains its magnetism without power supply. - Incorrect; an electromagnet requires a continuous electric current to maintain magnetic force, otherwise it loses magnetism when removed from the circuit (0 mark for this option as incorrect justification is not provided in brief).

(C) It generates heat that can melt metal objects nearby. - Incorrect; while some energy loss occurs due to resistance and heating within a coil, an electromagnet's primary function isn't generating enough heat to cause melting (0 mark for this option as incorrect justification is not provided in brief).

(D) It creates light that can illuminate metal objects. - Incorrect; while some types of magnetism are used in certain lights and displays, an electromagnet itself does not produce visible light when activated by a current through its coil (0 mark for this option as incorrect justification is not provided in brief).

Numerical: If the strength of magnetic force required to pull down a bell's clapper with 25 grams of steel wire attached increases linearly over time due to increasing tension, and it takes exactly 3 seconds from activation for this amount of weight to be pulled by an electromagnetic effect in ideal conditions (ignoring air resistance), calculate the average force exerted on the bell's clapper during these 3 seconds.

($F = ma$; $m = 0.025 \text{ kg}$, a is constant and cancels out) - Incorrect justification as it does not provide units for F or explain how to find ' a '. The correct answer cannot be calculated without additional information about the acceleration (if any). However, if we assume that there's no acceleration due

to gravity acting on the bell clapper in this scenario and only magnetic force is considered:

$F = mg$; where $g = 9.81 \text{ m/s}^2$ - Incorrect justification as it assumes gravitational pull which isn't relevant for an electromagnetic effect (0 mark since units are not provided). The correct answer cannot be calculated without additional information about the magnetic force constant or how to find 'F'. However, if we assume that there is a certain known strength of magnetism providing this specific pulling action:

(N/A; requires more data) - Correct as it acknowledges missing necessary details for calculation.

Q37. An electromagnet is created by wrapping a conductive wire around an iron core (such as a nail) and passing an electric current through this coil. When activated, it generates a magnetic field due to Ampère's circuital law which states that the integrated magnetic field along a closed loop surrounding a current-carrying conductor is directly proportional to the current flowing through the wire (Lenz's Law). This behavior can be advantageous in applications like electric doorbells or relays, where an electromagnet needs to break a circuit. When activated by sending sufficient current into it, its magnetic field becomes strong enough to attract and hold onto another piece of metal which is part of the mechanism that breaks the circuit when released (e.g., releasing arm). This allows for mechanical movement as well as control over an electrical connection without direct contact with conductive materials or complex machinery.

MCQ: When activated, what does an electromagnet do to a nearby metallic object?

A) Repels it (Correct - An electromagnet generates a magnetic field that can repel ferromagnetic objects when the current is reversed.)

B) Attracts and holds onto it due to its strong magnetic field. (Correct – When activated, an iron core within the coil becomes magnetized by the electric current flowing through it.)

C) Changes color of the metal object.

D) Makes the metallic object hotter without any physical contact.

Q38. A concave mirror can indeed produce both real and virtual images based on object placement. When an object is situated between the focal point (F) of a concave mirror and the vertex (V), where rays appear to diverge, it results in:

- A magnified erect image that cannot be projected onto a screen because its formation involves virtual light paths extending behind the mirror. This type of image is perceivable only by looking into the mirror' end as if there was an object located at this point (P).

- The ray diagram for such placement would show reflected rays diverging after reflection, and when extended backward, they seem to converge on a virtual focal point behind the mirror. This is where we perceive our magnified image of the real object placed between F and V. To illustrate this with an example:

1. Draw a concave mirror as a curved line (with its center being the vertex). Mark points for P, F, and R (the point on the principal axis directly opposite to the vertex) accordingly.

2. From object O located between V and F, draw two rays: one parallel to the principal axis reflecting through F, another passing through F before reaching mirror surface without deviation due to its nature of being focused at a focal point (F). These are known as P1 and P2 respectively.

3. Extend these reflected rays backward behind the mirror; they appear to intersect on an imaginary line that is extended straight from V through F, which gives us our virtual image I located between O and F at a position where we can see it when looking into the mirror' end (P). This point P represents how far away in space this object appears due to magnification.

4. The characteristics of such an image are that:

- It is upright or erect as there's no flipping over during reflection, and it maintains the same orientation as the original object (hence 'erect').

- Magnified because this virtual image appears larger than the actual object due to its specific placement between F and V. The magnification can be calculated using mirror formula $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$, where f is focal length, v is image distance (negative for a real image), u is object distance (positive if in front of the mirror).

- It cannot be projected onto a screen because it's virtual; only light paths that actually converge at one point can form such images.

5. This phenomenon occurs due to diverging reflected rays, which our brain interprets as coming from behind the mirror when extended backward (a trick of human perception). The image appears larger because it is formed by a wider range of angles between these virtual light paths compared to those forming real images at other object placements.

6. This type of imaging can be useful in various applications such as shaving or applying makeup, where magnification and uprightness are beneficial features provided the viewer is looking into a concave mirror's end (P).

In summary: When an object lies between F and V on a concave mirror surface, it produces a virtual image that appears erect, magnified, and located behind the mirror. This phenomenon occurs due to diverging reflected rays interpreted by our brain as coming from this point when extended backward (P). The characteristics of such an image are uprightness, magnification beyond actual object size, invisibility on a screen because it's virtual but perceivable through reflection at P.

MCQ:

What type of image is formed by placing the object between F and V?

A) Real Image (Correct option - real images are not upright or magnified; they appear smaller than actual objects.)

B) Virtual Image (Correct answer – virtual images cannot be projected onto a screen, appearing to form behind mirror due to diverging reflected rays. They maintain the same orientation as object and can be perceived by looking into the end of concave mirror.)